

Search sheets: how to run the activity



The pages after this one are 24 **search sheets**, built from *The Cat in the Hat* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

Each sheet holds 81–82 **entries**. An entry is a **next word** paired with the **previous word** that came immediately before it somewhere in the original text.

In **we do**, the boxed word is the previous word and *do* is the next word. Each word keeps the same colour wherever it appears, so scan by colour first and read second. With only 12 colours and far more words than that, a colour narrows the search rather than settling it — always check the word itself.

1962 tokens, 240 unique · 2.83 bits/token · perplexity 7.1 · branching factor 4.32

Running a round

1. **Call out the last word** of the text so far.
2. Everyone scans their own sheet for entries whose previous word matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that entry's next word.
4. A scribe writes the word on the board. That word becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

Worked example

Start anywhere. Pick any entry — say this one — and write its previous word *and* its next word on the board.

Round 1. Call out the last word on the board (*do*). Another sheet has a match, so that person raises a hand and reads out *not*.

Round 2. Call out the last word on the board (*not*). Another sheet has a match, so that person raises a hand and reads out *bite*.

Keep going for as long as you like. The text the room writes will sound like *The Cat in the Hat* without ever repeating it — that is the model generating, not the corpus being read out.

What to watch for

The hands are the probability distribution. Every entry was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Running dry. If no hands go up, you have reached a context that only ever occurred at the very end of the text. Start again from any entry.

the entry that answers

we do

do not

not bite

on the board

we do

we do not

we do not bite

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "we", then the entry **we** do is a match and you answer **do**.

we do	big bumps	then he	, wet
and so	wet to	, said	cup on
to you	sad kind	! so	little toy
show to	bump on	do you	the fish
and so	things will	fish said	picked up
like ,	Why do	my net	I can
should not	play .	. with	two and
said the	away !	they should	! her
up !	it came	cat in	in the
down !	tell me	the box	had to
, wet	on a	gone away	not bite
is not	the fish	. Oh	. it
fun on	milk on	fast as	? would
one little	! said	mother !	with the
a house	that day	her about	Thing two
you some	will show	be here	what would
the house	. have	no !	But that
how .	. that	net .	fan !
box are	and the	said .	in the
I did	have to	I do	! said
of good	Oh ,		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "they", then the entry **they** **should** is a match and you answer **should**.

dear .	they should	fish came	him !
with my	milk and	, said	we had
day .	on the	have Lots	! said
you do	wet day	will not	! do
fast !	do not	fun at	mess ,
down the	said the	I will	to us
in the	and Thing	one and	. a
, white	cat in	box .	that is
! said	But you	. your	and a
. this	her bed	the fish	mother will
, and	I do	things and	not like
fan as	. your	I call	what to
the gown	call up	be about	then he
little fish	said the	about it	cup and
show them	Why ,	up with	tall ,
like to	away .	the fish	net ,
play !	Oh dear	looked !	after my
a game	house and	? Oh	big thumps
two hands	my net	no .	it is
. a	of look	! said	is a
the things	and the		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "I", then the entry **I said** is a match and you answer **said**.

I said	. .	looked !	no !
play ,	deep and	said the	cup ,
big mess	mother asked	like it	cat .
. they	have any	and I	away !
her gown	will see	fun ?	about with
it !	But your	I like	the ship
. he	of his	sat in	the mat
call them	I can	sun did	up the
fan ,	box with	on a	not one
. we	wet ,	you can	one one
we saw	! have	him step	that cat
then our	! and	and red	back in
they will	the cat	fish .	to us
show you	house .	, here	things I
the top	and I	with a	dear !
top of	that they	! those	do was
, said	ball !	mess is	what will
down !	said the	. look	is not
day .	a good	! at	? and
my net	net I	, I	milk ,
in the	little bit		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "cup", then the entry **cup** **and** is a match and you answer **and**.

cup and	! look	fun But	I can
that they	them go	him pick	of Thing
on there	the sun	little bit	deep in
mess !	the cat	up two	with a
. then	said the	do you	in the
you ?	are pink	sat there	fan with
we did	and the	I did	shook their
the cat	and big	. with	my net
day .	and I	, said	bow .
mother will	she will	! then	if your
this mess	will like	like these	. two
top with	hit !	But I	! look
, what	no .	play ball	. that
one and	look at	things fall	away with
down .	, we	cat in	not wish
ball .	it is	to find	take them
call this	they want	! I	the hat
said ,	back in	is not	box .
sun is	what to	! how	I will
fish on	wet and	house !	a game
about .	. there		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "it", then the entry **it to** is a match and you answer **to**.

it to	. you	no !	and we
shook up	there like	did not	good fun
my hat	box came	. it	the sun
little bit	play .	a ball	in a
I can	. .	fall .	. I
do ?	we looked	! have	and the
wet day	said the	like it	they ran
day ?	our fish	I can	things they
if you	this mess	fish to	on his
a shame	of one	what a	, how
will show	take a	sat there	down ,
! and	with a	to us	! said
! then	them a	mother came	said the
kite bump	away .	the cat	not all
gown with	one hand	are two	the milk
cat .	the cat	ran up	, said
look !	fun that	house !	But that
When your	ball .	. .	that is
and the	hook .	she do	I saw
, as	lit .	! and	here .
you should	up the		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "we", then the entry **we** saw is a match and you answer **saw**.

.	we saw	it .	play .
. then	she say	ball .	our fish
I saw	here When	shook with	not .
things yet	, we	hook .	did nothing
ran down	I call	asked you	little toy
, Oh	. our	you hear	fun ,
they have	so ,	gown !	at me
sat in	do !	one !	, now
! I	the cat	there with	this game
I do	kite we	look at	if I
will not	what the	a PLOP	yet !
said the	are good	with Thing	cat .
When our	and down	. .	of the
and Thing	up the	hold up	the cat
that I	But our	! they	! then
mother is	to shake	good ,	he should
down with	not be	! I	a cup
the cat	. and	and she	! I
fall !	them out	new tricks	on a
the wall	like to	in the	house ?
now you	said the		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "always", then the entry **always** pick is a match and you answer **pick**.

! and	always pick	we saw	good games
this way	I said	the things	house !
and a	and this	of my	what did
mother is	did you	fun .	I can
down !	them out	said the	do not
is what	so deep	, and	the house
the hat	I know	ball !	fell on
. they	. I	fall !	! on
tame .	look at	at all	their two
will she	on the	gown ,	one !
to be	you sank	like it	they ran
our toy	she will	not like	the hat
are tame	ran to	head !	hold you
things in	cat .	hook .	he picked
if mother	there ,	, and	. I
Make that	new .	with a	But that
, little	and Thing	bit !	not all
a little	said ,	! now	in the
now look	! he	! and	a cup
that !	. they	it ?	here When
. and	When your		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "she", then the entry **she** say is a match and you answer **say**.

she say	new rake	! Oh	here is
I bet	cat with	you wish	they should
and with	did not	Make them	then we
! put	it up	do to	cake !
I will	bit !	good tricks	the things
like that	one little	. Oh	the fish
house !	he should	this house	, said
not like	mother is	the pot	and the
their kites	tame !	the cat	what to
and the	, wet	in the	on the
said the	. what	! you	, cold
said the	a bow	us two	ran about
. you	to get	know how	. now
a shame	not like	game that	! and
But we	. my	head of	is not
are not	! you	that is	. I
there that	of his	shut with	now !
hold up	so so	fell into	another good
things that	look at	at this	will hit
we had	with big	them I	I said
our mother	has gone		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "did", then the entry **did** not is a match and you answer **not**.

did not	. .	shut up	, no
hear !	. But	one little	the rake
things .	this trick	! they	cake .
red .	of good	now !	not like
and the	sit !	the hat	For the
I bet	saw all	, not	all the
not .	I did	! with	to fall
game fun	know what	, and	we saw
a pat	the things	another good	bit .
that ?	! those	said the	said the
us jump	then he	house !	. and
what you	hold a	on my	sank it
gave them	! tell	look !	you up
at all	cat ,	in the	good trick
is not	a fox	he should	. they
I will	like to	will have	But our
the things	ran out	so we	. I
do !	she said	the fish	them Thing
there on	and ,	! sit	with hops
they said	. you	new gown	cold to
our mother	here to		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "I", then the entry **I sat** is a match and you answer **sat**.

, as	I sat	will show	! not
on the	know !	game .	. what
to do	. she	said the	cold ,
. Why	! the	things .	, then
red wood	fox ,	they ran	we can
look what	. then	see this	! Make
I do	the hook	said ,	and then
them to	the cat	fish in	at me
us ?	here .	saw those	in the
of bad	did !	like this	he fell
a hook	the cat	sit !	cat .
and he	hold up	I do	what will
! bump	sank our	hear ?	all .
this wet	! Oh	not fly	can not
, Think	into a	the fish	with a
so tall	not be	you pack	do you
wish we	. they	then !	house all
a look	now !	. should	pack up
as I	were down	one kite	there is
that cold	is not	good game	cake ,
me !	our fish		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "not", then the entry **not** be is a match and you answer **be**.

not be	. .	wish to	in this
good things	said ,	here When	me .
on the	. and	now ,	a shame
! what	the hat	as I	to know
can ,	, the	did not	our fish
all kinds	what would	house For	like the
cat went	cold ,	the dots	sit !
you sit	he ran	books !	this box
game ,	at that	man !	to have
we saw	that cat	kinds of	way that
a rake	the strings	not go	fish .
will not	, she	I know	and some
, and	red fan	see something	hold up
look !	things ran	cake on	saw mother's
. then	! Oh	bent our	then he
know it	so do	. then	! I
the pot	. Oh	bumps ,	hat .
up and	! they	us fast	do if
give you	with a	is good	! and
one kite	I know	them away	and the
I hop	. now		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "and", then the entry **and** **the** is a match and you answer **the**.

and the	, we	! down	and then
we saw	game that	, if	books ,
the day	I can	on the	you like
all we	look at	Who was	said ,
he has	way !	mother's new	this is
can do	will give	what a	. I
things they	wood box	the dish	is near
. then	kites in	hold the	a book
! not	high as	. what	us ,
. I	the head	! and	I can
like it	not want	one and	now look
a lot	our house	with a	shine .
. and	then we	fish !	sit there
fly kites	know some	wish ,	not all
did not	do not	hat !	playthings and
, the	! they	! But	to the
you will	lot of	cat !	to play
so ,	in the	as fast	them away
and he	. these	saw her	at me
up these	that went	the cat	two things
me now	good fun		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "saw", then the entry **saw** **them** is a match and you answer **them**.

saw them	what a	find this	. look
cat .	. and	to say	us this
a tip	in and	! then	. and
do as	! look	sit !	the cat
way home	a pot	said ,	our new
is wet	! look	me !	nothing at
out of	these things	and kicks	I call
two books	fly kites	hat you	not mind
I will	game that	wish to	trick that
all !	things that	this !	, Why
books !	up the	, I	as a
! put	he said	, we	not be
look at	with a	pat .	the ball
you take	! Make	fish in	then I
. Oh	. this	the way	that is
the house	can have	at all	will not
, the	you another	lot !	and I
kites on	good game	them run	. not
to us	ship ,	now ,	know !
like our	so ,	tip of	and he
we sat	did not		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "with", then the entry **with** a is a match and you answer **a**.

. Oh	with a	these two	. I
this ?	did you	. .	! and
! and	way at	! But	is fun
the cup	up on	. so	! said
me down	find us	put them	I do
shame !	he put	no fear	last !
look !	will come	do something	ship !
at last	you hear	the milk	we can
in the	saw him	to play	, what
fish ,	. he	hat .	came down
cat in	, with	said the	then we
them !	the hat	Yes .	stop .
and the	I can	. But	stand on
trick ,	Thing one	now .	tricks .
all .	a cake	! fast	the hall
as I	pot !	, do	can hold
Oh ,	out and	two things	to the
you have	that is	, no	not all
so tame	tip of	and the	kites ,
not one	hands with	our mother	a box
and look	know what		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "as", then the entry **as** you is a match and you answer **you**.

as you	shame !	something went	Thing two
Oh ,	. you	he said	hands .
at me	the cat	! no	to go
the cake	, and	. .	then ,
. .	the books	, no	said the
, did	, I	funny !	! we
fish !	hop up	two ,	and I
hall .	let you	not wish	do !
me down	with the	hat .	rake .
! put	you do	know some	ship ,
! Thump	that are	out !	have to
cat .	tricks .	toy ship	our fish
and Sally	we sat	a bump	all !
the cat	not a	you shook	came in
. I	I said	saw him	. and
so so	will bump	up those	? Oh
pot .	this is	put me	and then
these things	now look	no ,	is that
mat !	look !	. have	can hold
bad tricks	should we	! Who	to say
I know	in a		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "hands", then the entry **hands** **with** is a match and you answer **with**.

rake ,	, with	hands with	! so
this house	you get	. that	all !
the cake	hop on	the fish	he fell
. Yes	with a	, I	tricks ,
our fish	! said	fish .	toy man
and you	the hat	and we	will find
at this	to do	you bent	out !
. I	shame !	a lot	, said
not know	? Well	. and	. I
is not	and so	ship and	then Sally
pot .	we sat	, the	! said
! then	want to	. do	do ?
look at	me !	I let	Thing two
so all	as I	book one	let down
should not	Oh dear	know what	hat .
put them	! and	saw one	now !
the wall	to stop	up up	bad ,
I will	can hold	cat in	these books
something to	not .	in the	that is
hall .	said the	came down	no fear
have no	two .		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "toy", then the entry **toy ship** is a match and you answer **ship**.

Oh,	toy ship	his hat	, with
this,	tricks are	run down	have no
Think of	no fun	. we	to play
hat to	at all	our game	not sunny
! and	not pick	with the	. you
do not	, no	! what	rake !
some good	fish in	and Sally	do not
out of	that bump	as he	he gave
came down	something new	the things	all .
put me	. they	. so	I do
your mother	then Sally	Thing one	I say
! too	is so	so .	and Sally
saw him	would you	and a	fear .
can hold	up high	? Why	should not
! I	to do	! now	want to
said to	we looked	you .	two things
, jumps	cat .	the toy	. Thing
any fun	in the	home !	, look
a big	look at	me now	get them
go !	sunny .	you will	the hook
pot .	the house		

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "those", then the entry **those** things is a match and you answer **things**.

those things	to the	the box	he said
? tell	not all	and I	toy ship
go away	two things	should not	this house
Oh ,	the cat	all the	so we
at me	with my	Thump !	kind of
! bump	put them	some fun	too .
you now	can hold	hat ,	bump !
pot .	, said	Sally .	as fast
up .	and Thing	fear !	his hat
do ?	came Thing	! and	, I
out !	said the	! sit	I always
! said	would she	the hall	fish as
. now	I wish	no fear	! look
the hat	then I	. no	have fun
Thing one	cat in	, and	get out
. I	something to	and a	. take
to fly	. But	not know	hand !
we tell	look .	in on	bed !
, and	Lots of	is on	a little
you do	From up	got up	that .
your mother			

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "should", then the entry **should** **not** is a match and you answer **not**.

should not	get those	two and	with a
I like	was gone	hat !	no fear
Oh ,	can hold	fear !	Sally and
. in	shake hands	! he	Thump !
Thing one	cat .	thumps and	came back
you do	the cat	they will	, and
, how	.	! then	at this
? and	not like	bump made	then those
all .	and they	look at	went on
those things	hops and	. it	up all
his head	pick up	you do	is out
the cat	your mother	some new	. But
the cat	not be	games we	in the
of her	, said	too wet	! put
! sit	he came	would you	.
to you	I had	the cat	so big
fish said	! I	a tip	this big
do .	we can	have no	that is
and I	and Thing	Lots of	out of
said the	Well .	we saw	go away
something !			

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "the", then the entry **the** ball is a match and you answer **ball**.

? I	, that	the ball	have Lots
no !	go out	! we	of the
cat in	mind at	we shook	then ,
should we	out .	was out	not know
and then	Sally and	do not	we could
to sit	Oh ,	so .	the fish
some milk	too cold	will hold	and the
all I	and you	jump !	up all
. .	those things	on top	! no
went away	he was	two !	bite you
! and	pick up	, too	that is
your mother	. he	it was	things had
. we	! then	with Sally	the cake
I will	. I	shake hands	say .
they will	wall !	be in	bump !
fear ,	! and	not be	I know
the fish	near !	a fish	. he
you fall	get rid	could see	fish said
in the	said the	, and	can get
is no	hat .	you did	said our
Thing one			

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "be", then the entry **be** here is a match and you answer **here**.

be here	and the	should we	will show
cat in	they play	we two	in the
I can	to have	.	the cat
not one	not shine	could ,	, said
can fan	he lit	a box	too wet
fish .	! they	then our	! But
, fast	your mother	jumps and	do not
have no	no fear	. you	go !
of this	. and	mother was	! have
the pot	? what	dots that	! said
my tricks	up the	said the	it ,
the fan	Thing two	things bump	two ?
is no	PLOP !	Sally and	and our
you have	said the	hat .	was to
with my	I stand	went bump	. But
tail I	dish !	Oh ,	that is
, said	. he	those two	you .
! then	the cat	wall in	so we
bump From	bet .	and the	down on
fear of	say .	all that	on her
pick it			

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "rid", then the entry **rid** of is a match and you answer **of**.

rid of	fish shook	you another	be here
it deep	could do	, out	made us
no !	tell her	fear .	, I
they like	had something	on the	? should
, sank	! Thump	cat .	say .
! Oh	things should	we did	to shake
said ,	my tail	have fun	should not
dish ,	in the	is not	down the
and I	not in	. I	.
that were	your mother	hat !	those two
bump !	her the	will pick	.
he got	of this	pick up	said the
Thing two	Sally and	was too	in this
! said	! and	I will	up there
went after	do .	a red	with fear
not let	. that	and so	. your
can get	then he	two things	come here
the hat	all if	! it	mother could
I like	the fish	the fan	and too
how do	fast as	the cat	bet ,
Oh ,			

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "be", then the entry **be here** is a match and you answer **here**.

be here	we could	up our	no !
your mother	! I	my playthings	cat go
fast as	will she	. he	, what
say ?	do you	white and	he shut
fish .	. I	all my	could play
Sally and	kicks and	Thing two	? Oh
tell her	. so	! I	big red
not !	! and	said to	in the
down with	the house	how that	I can
it a	string of	step in	of good
and our	is funny	with a	in the
then something	two .	to play	wet to
have no	come .	mother is	those things
Oh ,	hat came	a dish	! your
was shut	, said	I went	had them
should not	. we	said ,	bump their
and I	her .	. then	that I
on this	they like	the hat	the string
the cat	, no	fun to	and all
cat in	you did	things are	! look
can hop			

When the word called out matches the boxed word of an entry below, raise your hand, and read out that entry's **next word** if you're picked. If more than one entry matches, just go with the first one you spot. For example, if the word called out is "tell", then the entry **tell** **that** is a match and you answer **that**.

pink ,	tell that	the cup	, said
have come	be here	how I	Thing one
then he	! now	? now	like it
big and	it was	away .	cat said
of something	said ,	and I	what should
on the	they are	box ,	I ,
had to	net .	. these	things ,
can have	the fish	do I	wet ,
hat .	one and	you will	! so
is not	a sad	. and	and look
! said	down my	! if	that I
the ball	gone with	in with	. you
in the	show you	I could	cat in
. no	with Thing	the pot	mother is
Why ,	was back	your mother	fast .
and Thing	, so	not bad	Oh ,
said to	to go	fish in	two .
strings ,	. so	up up	should not
, so	fun in	say !	her way
bump !	no way	my hat	will not
we tell			